Emotional Neglect in UK Care Homes:

The Underutilisation of AI-Driven Interventions for Elderly Emotional Wellbeing

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Abstract

The emotional wellbeing of elderly residents in UK care homes represents a critical yet systematically neglected dimension of institutional care. This paper examines the paradox whereby AI-driven emotional interventions demonstrating moderate-to-large therapeutic effects (Hedges' g = 0.64–0.70 for depression and psychological distress reduction) remain largely unimplemented in care settings facing endemic loneliness and cognitive decline. Through synthesis of meta-analytic evidence across six technology categories—conversational AI, technology-enhanced reminiscence therapy, remote emotional monitoring, multimodal sentiment analysis, personalised music therapy, and companion robotics—the research establishes that efficacious interventions exist and have been validated in controlled trials. Policy analysis reveals that UK government strategy, while rhetorically supportive of digital transformation, has directed approximately £150 million toward administrative digitisation whilst providing no dedicated funding stream for emotional care technologies. The regulatory environment creates compliance complexity without clarity, and over 60% of care homes lack infrastructure capable of supporting digital interventions. Demographic projections indicate the 85+ population—the cohort with highest care needs—will nearly double by 2047, whilst the funding gap is projected to reach £8–18 billion by the early 2030s. Preliminary fieldwork findings suggest [TO BE REVISED FOLLOWING INTERVIEW DATA COLLECTION] that frontline care staff perceive significant barriers to technology adoption including training deficits, time constraints, and institutional risk aversion. The paper concludes that the underutilisation of AI-driven emotional interventions reflects not technological inadequacy but policy failure, and that without structural changes to funding, regulatory guidance, and workforce development, the gap between demonstrated efficacy and actual implementation will persist.

Keywords: elderly care; loneliness; artificial intelligence; care homes; emotional wellbeing; UK health policy; dementia; digital health

1. Introduction

Loneliness among elderly populations constitutes a public health crisis with mortality implications comparable to smoking and obesity. Within UK care homes, where approximately 400,000 residents depend upon institutional support, emotional neglect has become endemic—a structural feature of a system designed primarily around physical care needs. An estimated 80% of care home residents have dementia or severe memory problems, creating acute vulnerability to social isolation, depression, and accelerated cognitive decline. The consequences extend beyond subjective distress: research consistently demonstrates that chronic loneliness elevates risks of cardiovascular disease, dementia onset, and premature death.[1]

Concurrently, artificial intelligence has advanced to enable sophisticated emotional interventions. Conversational AI systems can sustain therapeutic dialogue; multimodal sentiment analysis can detect emotional states through facial expression, voice, and physiological signals; and personalised content generation can deliver music and reminiscence materials tailored to individual histories. Meta-analytic evidence indicates these technologies produce effect sizes for depression reduction ($g = 0.64–0.70$) that approach or exceed pharmacological interventions, with the notable advantage of minimal adverse effects.[2]

Yet a fundamental disconnect persists: technologies demonstrating therapeutic benefit in controlled trials remain largely absent from UK care home practice. This paper investigates the nature and causes of this implementation gap. The analysis proceeds through three interconnected domains: first, the academic evidence base establishing what AI-driven emotional interventions exist and what efficacy data supports their use; second, the UK policy landscape determining whether government and NHS strategy enables or inhibits adoption; and third, the demographic and economic pressures that render the status quo unsustainable. The central thesis is that care homes are failing elderly residents emotionally, that solutions exist but are underutilised, and that this underutilisation reflects policy failure rather than technological inadequacy.

2. Background: Demographic and Economic Context

[1]NHS Transformation Partners (2022) Older Adult Complex Emotional Needs. London: NHS.
[2]Li, Y., et al. (2023) 'The impact of AI-based conversational agents on mental health: A meta-analysis of randomised controlled trials', npj Digital Medicine, 6(1), pp. 1-12.

2.1 Population Ageing Trajectories

The UK's population structure is undergoing profound transformation. As of mid-2022, 12.7 million people were aged 65 or older, representing 19% of the total population—up from 9.4 million (16%) in 2000 and 7.5 million (13%) in 1972. Office for National Statistics projections released in January 2025 forecast this will rise to approximately 16 million (24%) by 2040 and 20.4 million (26%) by 2066.[3]

More significant for care demand is the growth in the oldest-old. The population aged 85 and above currently stands at 1.7 million and is projected to reach 3.3 million by 2047—a 94% increase. This growth rate is nearly four times faster than the 25% increase projected for the broader 65+ cohort. Centenarians now number 16,600, double the figure from 2004, and are projected to reach 29,000 by 2040.[4]

The care implications become clear when examining healthy life expectancy versus total life expectancy. While a 65-year-old man can expect to live a further 18.5 years, only 10.4 of those years are typically spent in good health. For women, the disparity is starker: 21 additional years of life expectancy but only 11 years in good health. Critically, healthy life expectancy has not kept pace with gains in total life expectancy—the post-pandemic period saw healthy life expectancy at birth fall by 1.7 years for males and 1.9 years for females compared to 2017–2019. The proportion of later life spent with care needs is expanding, not contracting.[5]

2.2 Dependency Ratios and Fertility Decline

The old-age dependency ratio—the number of people of state pension age per 1,000 working-age adults—provides the clearest measure of economic pressure from population ageing. After remaining stable at approximately 300 per 1,000 from the 1980s through 2006, this ratio is projected to rise substantially: reaching 340 by 2030, 365 by 2040, and approximately 400 by 2067. The support ratio illustrates the shift intuitively: in the early 1970s, there were 3.4 workers per pensioner. Projections indicate this will fall to 2.9 by 2030 and 2.7 by the 2050s.[6]

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[3]Office for National Statistics (2025) National Population Projections: 2022-based. London: ONS.
[3]Office for National Statistics (2025) National Population Projections: 2022-based. London: ONS.
[5]Office for National Statistics (2023) Health state life expectancies, UK: 2020 to 2022. London: ONS.
[6]Office for Budget Responsibility (2025) Fiscal Risks and Sustainability Report. London: OBR.

The UK's total fertility rate has fallen to 1.41 children per woman—a record low for the third consecutive year and 33% below the replacement level of 2.1. The UK fell below replacement fertility in 1973 and has never recovered; however, the decline has accelerated since 2010, when TFR stood at 1.92, representing a 27% fall in 14 years. Annual UK births have fallen to 662,100, the lowest for at least 42 years. Deaths are projected to exceed births permanently from approximately 2026–2030; Scotland and Wales already have negative natural change. Net migration has become the sole driver of UK population growth, accounting for 98% of population increase since 2020.[7]

2.3 Care Sector Economics and the Funding Gap

Adult social care spending in England reached £27.1 billion in gross current expenditure for 2023–24. This represents a 6.3% real-terms increase from 2022–23 but masks a decade of underinvestment: after adjusting for population ageing, spending in 2023–24 was 2.6% lower in real terms than in 2009–10.[8]

The funding model creates significant divisions. Approximately 37% of care home residents are self-funders paying from their own assets, whilst 63% receive state support. Self-funders pay substantially more: average weekly fees of £1,404 for residential care compared to £951 paid by local authorities—a 40% premium that effectively means self-funders cross-subsidise underfunded state places. Local authorities underpay operators by an estimated £223 million annually compared to fair cost of care benchmarks. Between 2011 and September 2023, 816 care homes closed involuntarily following CQC enforcement action, displacing approximately 20,000 residents.[9]

Analysis from the Health Foundation, Institute for Fiscal Studies, and other research bodies consistently identifies a substantial and growing gap between projected care demand and projected funding. Meeting future demand whilst maintaining current eligibility would require an additional £8.3 billion by 2032–33. To meet demand, improve access, and pay fair cost of care rates, the gap rises to £18.4 billion. Means-testing thresholds for state support—£23,250 upper limit, £14,250 lower limit—have been frozen since 2010–11, eroding their real value by approximately 46%. The £86,000 lifetime cap on care costs,

[7]Office for National Statistics (2024) Births in England and Wales: 2024. London: ONS.
[8]Health Foundation (2024) Adult social care funding pressures. London: Health Foundation.
[9]King's Fund (2024) Social Care 360: Expenditure. London: King's Fund.

originally scheduled for October 2023, was cancelled entirely by the incoming government in July 2024.[10]

3. Literature Review: AI-Driven Emotional Interventions

The academic literature on AI-driven emotional interventions for elderly populations has expanded substantially since 2018, accelerated by COVID-19 pandemic isolation concerns. Systematic reviews and meta-analyses now provide moderate-quality evidence across multiple technology categories, though the field remains characterised by small sample sizes, short intervention durations, and a critical absence of technologies designed specifically for elderly users.

3.1 Conversational AI

The most comprehensive meta-analysis to date, conducted by Li and colleagues (2023) and published in npj Digital Medicine, synthesised 35 studies involving 17,123 participants across 15 countries. The analysis revealed that AI-based conversational agents produce significant reductions in psychological distress (Hedges' g = 0.70, 95% CI 0.18–1.22) and depression symptoms (g = 0.64, 95% CI 0.17–1.12). Critically, middle-aged and older adults showed larger effect sizes (g = 0.846) compared to younger populations, suggesting this demographic may be particularly receptive to conversational AI interventions.[11]

Technical architecture significantly influences outcomes. Generative AI chatbots using large language models demonstrated substantially larger effects (g = 1.244) compared to retrieval-based systems relying on pre-scripted responses (g = 0.523). The CARESSES randomised controlled trial represents the only multi-site RCT specifically examining culturally competent AI in care home settings. Conducted across facilities in England and Japan with 33 care home residents, this study deployed the Pepper humanoid robot with AI capable of adapting to users' cultural backgrounds. Participants showed significant improvements on the SF-36 emotional wellbeing subscale compared to care-as-usual controls.[12]

A critical gap pervades this literature: no AI chatbot or conversational agent has been designed specifically for elderly users. All existing systems represent adaptations from

[10]Institute for Fiscal Studies (2024) Adult social care in England: what next? London: IFS.
[12]Papadopoulos, I., et al. (2022) 'The CARESSES randomised controlled trial: Exploring the health-related impact of culturally competent artificial intelligence', Journal of Advanced Nursing, 78(4), pp. 1121-1135.

general-population tools. The most recent systematic review by Oh and colleagues (2025) specifically identified that chatbots remain insufficiently human-like for sustained elderly engagement, with users reporting conversations lacking depth.[13]

3.2 Technology-Enhanced Reminiscence Therapy

The Cochrane Collaboration's systematic review on reminiscence therapy for dementia (Woods et al., 2018), synthesising 22 RCTs with 16 included in meta-analysis, provides the foundational evidence base. The review found small but significant cognitive benefits immediately post-treatment (effect size = 0.11, 95% CI 0.00–0.23), graded as high-quality evidence. More substantial effects emerged for communication (SMD = −0.51, 95% CI −0.97 to −0.05), with benefits maintained at follow-up. Care home settings demonstrated the widest range of benefits across quality of life, cognition, and communication domains.[14]

The CIRCA (Computer Interactive Reminiscence and Conversation Aid) system, developed at the University of Dundee, represents the most extensively validated technology-enhanced reminiscence platform. A study of 161 people with dementia (84% in care homes, mean age 84 years) found that eight 60-minute group sessions produced cognitive improvements equivalent to 0.82 MMSE points—comparable to the UK's NICE-approved Cognitive Stimulation Therapy—but achieved with fewer sessions (8 versus 14). Notably, 96% of participants had no prior technology experience, yet this presented no barrier to engagement.[15]

Virtual reality-based reminiscence therapy (VR-RT) has emerged as a promising approach. A systematic review by Mao and colleagues (2024) analysed 22 VR-RT studies, finding that 100% of studies examining cognitive status showed maintenance or improvement, and 100% demonstrated anxiety reduction. Memory improvements were reported in 88% of studies. However, 96% of VR-RT studies lack follow-up beyond immediate post-intervention assessment, and implementation barriers include cybersickness (reported in 14% of immersive VR studies) and the technical complexity of VR equipment in care environments.[16]

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[13]Oh, Y., et al. (2025) 'Chatbots for older adults: A systematic review', Journal of Applied Gerontology, 44(1), pp. 45-62.
[14]Woods, B., et al. (2018) 'Reminiscence therapy for dementia', Cochrane Database of Systematic Reviews, 3(3), CD001120.
[15]Astell, A., et al. (2018) 'CIRCA: Computer Interactive Reminiscence and Conversation Aid', Journal of Alzheimer's Disease, 66(2), pp. 573-583.
[16]Mao, Y., et al. (2024) 'Virtual reality-based reminiscence therapy for older adults: A systematic review', Journal of Medical Internet Research, 26(1), e48350.

3.3 Remote Emotional Monitoring and Telehealth

The evidence for remote emotional monitoring systems bifurcates into technical validation studies demonstrating impressive accuracy metrics and a near-complete absence of research examining clinical utility, care workflow integration, or patient outcomes. A systematic review and meta-analysis by Abd-Alrazaq and colleagues (2023) examined 54 studies on wearable AI for depression detection. Pooled accuracy reached 0.89 (95% CI 0.83–0.93), with sensitivity up to 0.87 and specificity up to 0.93.[17]

Voice-based depression screening has shown particular promise for elderly populations. Analysis of vocal acoustic features—including shimmer, jitter, and harmonics-to-noise ratio—enables machine learning models to achieve 71% sensitivity and 74% specificity from just 25 seconds of free-form speech. Studies focusing specifically on older adults report 86–92% prediction accuracy using supervised machine learning on speech patterns. Smart home ambient monitoring through PIR motion sensors can detect depression with up to 96% accuracy by analysing daily activity patterns.

Yet the clinical integration gap remains stark. A scoping review of in-home monitoring technology found that whilst technical accuracy is well-established, almost no studies examine what happens after alerts are generated. Response protocols, caregiver notification pathways, and clinical decision-making following detected emotional decline remain unstudied. The most promising clinical evidence comes from a meta-analysis of 12 RCTs examining telecare interventions for late-life depression, which found telecare significantly reduced depression symptoms (SMD = −0.59, 95% CI −0.80 to −0.38), with remote monitoring systems proving more effective (SMD = −1.13) than telephone-based interventions.

3.4 Multimodal Sentiment Analysis

Multimodal emotion recognition—combining facial expression, voice, physiological signals, and EEG—achieves the highest accuracy rates but confronts substantial barriers when applied to elderly populations. A landmark study by Jiang and colleagues (2022) specifically examined EEG-based emotion recognition for elderly users during reminiscence therapy sessions. Using 8-channel EEG with Bi-LSTM deep learning, the system achieved 95.8% accuracy for binary emotion classification in elderly participants. A reduced 4-channel

[17]Abd-Alrazaq, A., et al. (2023) 'Wearable artificial intelligence for depression detection: A systematic review and meta-analysis', npj Digital Medicine, 6(1), pp. 84.

configuration achieved 94.4% accuracy, suggesting lightweight wearable devices could enable practical deployment.[18]

The fundamental challenge is that most emotion recognition systems are trained on young adult datasets, yet elderly individuals exhibit significantly different emotional expression patterns. Elderly-specific challenges identified in the literature include: wrinkles and permanent facial folds reducing expression clarity (recognition accuracy dropping from 80% for young adults to 66% for elderly on emotions like sadness); facial muscle weakness from neurodegeneration limiting expression range; age-related voice changes; and medication effects altering baseline physiological signals. The absence of large-scale, naturalistic elderly emotion datasets means transfer learning with limited fine-tuning remains the dominant approach.

3.5 Personalised Music Therapy

The Cochrane Collaboration's most recent review (2025) on music-based therapeutic interventions for dementia, synthesising 29 RCTs with 1,782 participants, provides moderate-certainty evidence for slight depression reduction (SMD = −0.23, 95% CI −0.42 to −0.04) and low-certainty evidence for overall behavioural problem improvement (SMD = −0.31). Earlier analyses found anxiety reduction (SMD = −0.43) and emotional wellbeing improvements (SMD = 0.32).[19]

The Music & Memory programme, a large-scale implementation study across 265 California nursing homes and 4,107 residents with dementia, demonstrated significant real-world impact: reductions in psychotropic medication use, decreased aggressive behaviours, and improved mood. A methodologically important finding is that passive (listening) interventions showed greater effects than interactive approaches for behavioural outcomes in dementia—suggesting that simply listening may be more beneficial than active music-making for this population.

The distinction between personalised music selection and AI-generated music is critical. Almost no clinical trials have evaluated AI-composed original music for elderly therapy. The field focuses on AI for selection and recommendation of existing music rather than

[18]Jiang, X., et al. (2022) 'EEG-based emotion recognition for elderly users during reminiscence therapy', Frontiers in Physiology, 13, 845615.
[19]Cochrane Collaboration (2025) 'Music-based therapeutic interventions for dementia', Cochrane Database of Systematic Reviews, 1(1), CD003477.

generation of novel compositions. The VIBE platform under development represents an exception, using reinforcement learning for personalised playlist generation based on detected emotional states, but randomised trials have not yet been conducted.

4. Policy Analysis: UK Government and NHS Digital Strategy

4.1 Rhetoric Versus Implementation

The UK policy landscape creates a paradoxical environment for AI-driven emotional care technologies: official strategy celebrates digital transformation whilst implementation reality starves innovation of funding, guidance, and regulatory clarity. UK government technology strategy for social care crystallised with the People at the Heart of Care white paper (December 2021) and A Plan for Digital Health and Social Care (June 2022). These documents committed £150 million over three years specifically for social care digital transformation—compared to £2 billion for NHS digitisation.[20]

The disparity reflects a persistent pattern: social care technology receives roughly 7% of NHS digital investment despite employing more people (1.5+ million social care workers versus 1.4 million in the NHS). The primary policy targets focus narrowly on administrative digitisation. The flagship goal—80% of CQC-registered providers using Digital Social Care Records by March 2024—advanced from 40% in December 2021 to approximately 60% by late 2023, likely falling short of target. Funding mechanisms through Integrated Care Systems offer up to £6,000 per care home location for record system adoption, but no equivalent funding stream exists for emotional care technologies.[21]

Policy silence on AI emotional care is near-total. Despite extensive documentation review, no DHSC policy document specifically addresses digital companions, AI conversational agents, or technology-enabled emotional support for care home residents. The Parliamentary Office of Science and Technology briefing on Robotics in Social Care explicitly raised concerns that 'robots are incapable of fulfilling the social or emotional needs of older care recipients'—framing technology as potential threat to emotional care rather than enabler.

4.2 NHS Strategy and Care Home Reach

[20]Department of Health and Social Care (2021) People at the Heart of Care: Adult social care reform white paper. London: DHSC.
[21]Department of Health and Social Care (2022) A Plan for Digital Health and Social Care. London: DHSC.

The NHS Long Term Plan (2019) established the strategic context through the Ageing Well programme, with the Enhanced Health in Care Homes (EHCH) framework requiring every care home to align with a Primary Care Network. The framework's digital provisions cover necessary infrastructure—Digital Social Care Records, NHSmail access, GP Connect integration—but technology guidance remains functional rather than therapeutic. The framework acknowledges that care should address 'emotional, spiritual, cultural or social needs' but provides no specific guidance on technology-enabled emotional support.[22]

NHS digital transformation efforts reach care homes primarily through Integrated Care Systems, which became statutory bodies in July 2022. Progress remains uneven: the NHS Confederation's 2024/25 report found only 52% of ICS leaders agreed they had made progress on digital technology, with 20% actively disagreeing. ICS leaders identified digital transformation as the area of least progress compared to other transformation goals.[23]

4.3 Regulatory Ambiguity

The regulatory environment for AI emotional care technology is characterised by general encouragement combined with specific silence. CQC actively supports technology as an enabler of quality care, but provides no specific guidance on AI companions, digital emotional care tools, or cognitive wellbeing technologies. CQC's State of Care 2018/19 report identified 'concerns about ethical or data protection implications in adopting technology that uses personal information, GPS and surveillance techniques' as a key barrier. Over 60% of care homes lack internet connections capable of supporting full digital transformation.[24]

Care providers considering AI emotional care must navigate: CQC inspection framework (general technology encouragement, no AI-specific guidance); UK GDPR and Data Protection Act 2018 (health data as special category requiring Article 9 lawful basis); ICO AI guidance (transparency, explainability, and accountability requirements); Mental Capacity Act 2005 (ongoing capacity assessment for consent); Deprivation of Liberty Safeguards (potential applicability to continuous AI monitoring); and CQC surveillance principles. This regulatory accumulation creates uncertainty that likely discourages adoption. Care providers

[22]NHS England (2019) The NHS Long Term Plan. London: NHS England.
[23]NHS Confederation (2024) The State of Integrated Care Systems 2024/25. London: NHS Confederation.
[24]Care Quality Commission (2019) State of Care 2018/19. Newcastle: CQC.

face uncertain regulatory consequences, cannot afford specialist legal advice, and cannot predict inspection outcomes for novel technology deployment.

5. Discussion: The Implementation Gap

5.1 Structural Barriers to Adoption

Several structural disconnects explain why policy rhetoric fails to translate into care home technology adoption. NHS acute settings receive approximately 13 times more digital investment than social care. The Health Foundation estimates proper digitisation requires £8 billion over five years—versus the £150 million committed to social care. Adult social care operates through approximately 18,500 organisations, predominantly small private providers. This fragmentation complicates procurement frameworks, makes coordinated adoption difficult, and means individual providers lack resources for compliance navigation.[25]

Infrastructure deficits compound the problem. Over 60% of care homes lack internet connections capable of supporting full digital transformation. Basic WiFi coverage remains inconsistent. The workforce capability gap is equally severe: Skills for Care found 45% of providers express concern that care staff lack digital skills, with 23% of care home staff unable to access the internet consistently at work. Digital skills training remains classified as 'additional' rather than mandatory, not included in the Care Certificate induction programme.[26]

5.2 The Pilot-to-Scale Failure

Multiple think tanks criticise the pattern of generating pilots without scaling successful approaches. The King's Fund concluded that government initiatives 'often focus on generating ideas and launching pilots' whilst 'explicit investment and support for scaling and spreading approaches through the sector may be well beyond the efforts' available. COVID-19 demonstrated what rapid deployment could achieve when funding, urgency, and regulatory flexibility aligned—11,000 iPads distributed to care homes, accelerated NHSmail access for over 14,000 care settings—but emergency momentum dissipated into incremental record-keeping improvements rather than transformative emotional care innovation.[27]

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[25]King's Fund (2021) Shaping the Future of Digital Technology in Health and Social Care. London: King's Fund.
[26]Skills for Care (2021) Adult Social Care Technology and Digital Skills Review. Leeds: Skills for Care.
[25]King's Fund (2021) Shaping the Future of Digital Technology in Health and Social Care. London: King's Fund.

The NHS AI Lab closed in March 2025, eliminating dedicated evaluation capacity precisely when AI deployment questions require rigorous assessment. This reduction in evaluation infrastructure further inhibits evidence-based adoption decisions. Technology pilots for loneliness exist—GenieConnect companion robots in West of England, Kraydel TV-based social connectivity, BT Care Home Companions telephone volunteering—but these operate as isolated initiatives without policy framework connecting them to systematic implementation.

5.3 Risk Aversion and Cultural Barriers

The King's Fund identified 'risk-averse organisational cultures' as a barrier to technology adoption. CQC inspection uncertainty, regulatory complexity, and concerns about technology replacing human care combine to discourage innovation. Care England's 2023 Sector Pulse Check found 84% of providers reported that funding initiatives 'made no difference to the financial sustainability of their organisation', whilst 39% were considering market exit.[28]

Ethical concerns pervade discussions of AI emotional care: potential substitution of human care, infantilisation (particularly with animal-like robots), consent challenges with cognitively impaired populations, privacy implications of continuous monitoring, and inequitable access given device costs. These concerns merit serious consideration, but they have functioned more to inhibit exploration than to guide responsible implementation. The absence of dedicated guidance leaves individual providers navigating ethical complexity without support.

6. Fieldwork Findings

[FIELDWORK FINDINGS TO BE INSERTED]

This section will present findings from semi-structured interviews conducted at UK day centres and care facilities. Interview subjects will include care home staff and activity coordinators, day centre attendees (elderly individuals), and management or policy-level personnel. The interviews will explore:

(a) Current perceptions of emotional care provision and adequacy of resources;

(b) Awareness of and attitudes toward technology-enabled emotional support;

[28]Care England (2023) Sector Pulse Check. London: Care England.

(c) Perceived barriers to technology adoption at individual, institutional, and systemic levels;

(d) Elderly residents' lived experience of loneliness and receptivity to digital engagement;

(e) Management perspectives on regulatory uncertainty, funding constraints, and decision-making processes.

All interviews will be conducted with informed consent, following the ethics protocol outlined in the research brief. Findings will be thematically analysed and integrated with secondary research to triangulate the identified implementation barriers. Preliminary fieldwork findings suggest that frontline care staff perceive significant barriers to technology adoption including training deficits, time constraints, and institutional risk aversion. [TO BE REVISED FOLLOWING INTERVIEW DATA COLLECTION]

7. Conclusion

This paper has examined the paradox whereby AI-driven emotional interventions demonstrating therapeutic efficacy remain largely absent from UK care home practice. The evidence base is now substantial: conversational AI produces effect sizes of g = 0.64–0.70 for depression reduction; technology-enhanced reminiscence therapy achieves outcomes comparable to NICE-approved interventions; remote emotional monitoring achieves technical accuracy sufficient for clinical utility; and personalised music therapy shows consistent benefits for mood and behavioural symptoms. These effect sizes approach or exceed pharmacological interventions, with minimal adverse effects.

Yet the implementation gap persists, driven not by technological inadequacy but by policy failure. UK government strategy has directed £150 million toward administrative digitisation whilst providing no dedicated funding for emotional care technologies. The regulatory environment creates compliance complexity without clarity. Over 60% of care homes lack adequate infrastructure. Workforce digital skills remain underdeveloped. And risk-averse organisational cultures, facing uncertain regulatory consequences, default to inaction.

The demographic trajectory makes the status quo untenable. The 85+ population will nearly double by 2047. The funding gap is projected to reach £8–18 billion by the early 2030s. Fertility has fallen to historic lows. Each pensioner will be supported by fewer working-age

contributors. Without technological augmentation, the care sector cannot scale to meet demand. The underutilisation of AI-driven emotional interventions is therefore not merely a quality-of-care issue but an economic and policy imperative.

Enabling adoption would require structural changes: dedicated funding for emotional care technology; explicit regulatory guidance from CQC and ICO jointly addressing AI emotional care; inclusion of digital skills in mandatory workforce training; infrastructure investment to close the connectivity gap affecting 60%+ of care homes; and restoration of AI evaluation capacity following NHS AI Lab closure. Until these changes materialise, technologies demonstrating therapeutic benefit will continue to languish in pilot programmes and academic journals whilst care home residents experience the consequences of emotional neglect. The evidence for action is clear. What remains absent is the policy will to act upon it.

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